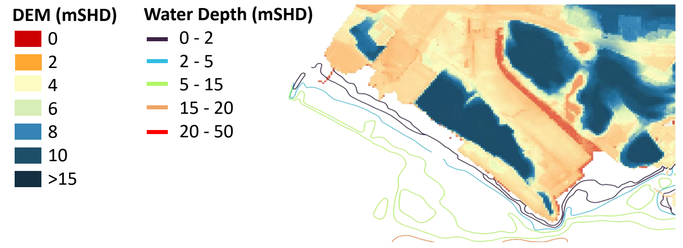
TRIFOLIATE HORIZON

CONNECTING MARINE, NATURE & PEOPLE

The concept is a picturesque rustic coastline that blends seamlessly with nature, showcasing a breathtaking seascape and lush greenery. Urban living is seamlessly integrated with the natural setting and water at Labrador Park and Berlayer Creek, bridging the gap between the green and blue. Residents can now experience nature up close, partake in various leisure activities, and indulge in our biophilic City in a Garden through a network of connectors (bicycle lanes and pedestrian pathways).



Labrador Nature Reserve remains with a redevelopment plan for a coastal park trail by Keppel. The land surrounding Berlayer Creek has over 6000 BTO Flats to be built to accommodate the growing population. The area is easily accessible through Labrador Park and Telok Blangah MRT.



Labrador Park and Pasir Panjang have varied low-lying levels between 2.5 to 3.0m SHD, except for the coastal forest area with levels above 8m SHD. Labrador Park's west side lies low at ~2.0m SHD. The northern shore of Berlayer Creek has relatively high terrain levels of 4.0 to 10.0mSHD, while at the southern shore, the land is significantly lower around +2.5-3.0mSHD. The focus of this project will be areas lower than 4.0mSHD.

LIVING SEAWALL

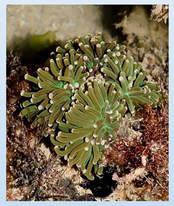
Sloping seawall defense is employed to maximize surface area, which encourages coral colonization and minimizes coastal flooding brought on by sea level rise.



Crinkled sandpaper coral (Psammocora sp.)

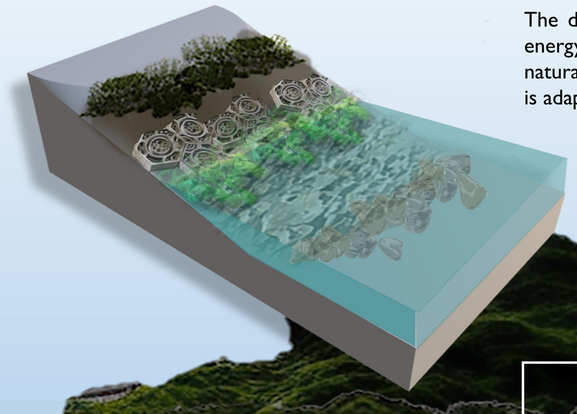


Pore coral (Porites sp.)



Anemone coral (Goniopora sp.)

The regenerated coral community on the seawall contributes to wave attenuation by acting as a natural barrier. It potentially grows with sea level rise and presents a source of blue cartons.



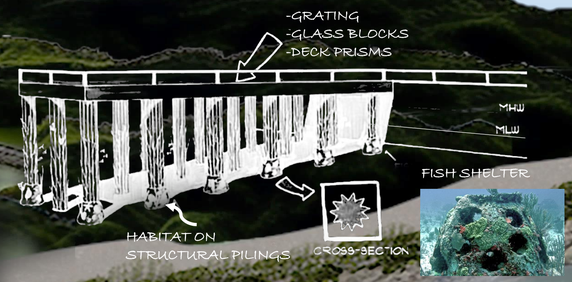
The design features a curved, sloping seawall that reduces wave energy and erosion by redirecting the wave upward and facilitating natural sediment circulation. When the water level reaches 5m, it is adaptable for modifications by increasing the crest level.

Bioactive concrete provides an organic substrate for coral growth with nooks, crannies, and fissures resembling natural reefs. It is durable in the marine environment and more sustainable with a reduction in Global Warming Potential (GWP) reaching 41% compared with traditional concrete.

ECOLOGICAL JETTY

An ecological jetty is an ideal substitute to replace the existing jetty which is currently situated at a low elevation.

The textured pilings and fish shelters create microhabitats by simulating natural environments. Fiberglass grating and reflective panels allow the high light intensity to pass through the dock, promoting photosynthesis and aquatic flora growth.



BIODIVERSITY ENHANCEMENT TILES & VEGETATIVE PLANTING

Biodiversity enhancement tiles and vegetation are strategically placed on the seawalls to enhance ecological features.



Biodiversity-enhancing tiles, designed like mangroves, offer diverse pits and holes that function as refugia, supporting a larger native intertidal wildlife population for example, purple climber crab (Metopograpsus sp.). Retrofitting these tiles enhances the shoreline's ecological value while preserving its integrity, unlike simple seawalls.

Introducing native coastal vegetation to the seawall's upper segment stabilizes the structure, reduces erosion, and enhances terrestrial wildlife habitats. It also enhances the waterfront's aesthetics, offers recreational opportunities, and fosters interaction with the coastal environment.



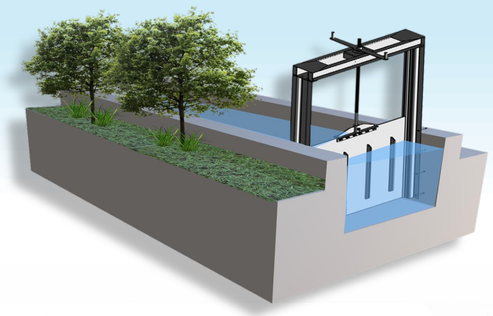
The jetty's extended platform offers leisure activities like cycling and fishing and brings people closer to nature. It attracts eco-tourists interested in discovering marine life in subtidal barrier reefs and regenerated rocky coastal habitats.

TIDAL GATE & PUMP

Consider building a tidal gate and pumping station upstream to minimize inland flooding during high-tide situations.

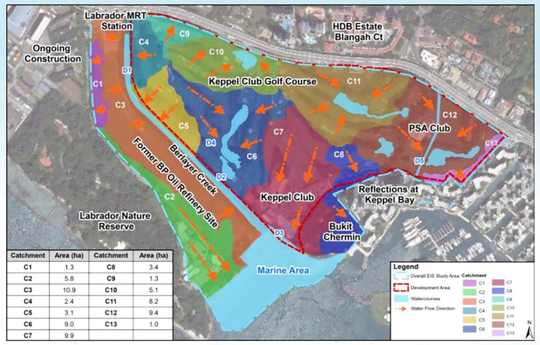
Sluice gates manage water levels and flow rates downstream. The gate is a component that can be lifted and lowered to block water flow. It is supported by a guide frame anchored into concrete and creates a watertight seal while sliding up and down.

The sluice gate can be controlled remotely with the aid of sensors positioned along the creek. To prevent inland flooding, the frequency of closing and pumping capacity must be decided upon when the water level reaches 5m.



A pump which is located near the base of the sluice gate is used to draw out excess water upstream. When the gate descends, the pump is immersed and begins to operate. Adding or enlarging the gate requires building work and has medium to high flexibility.

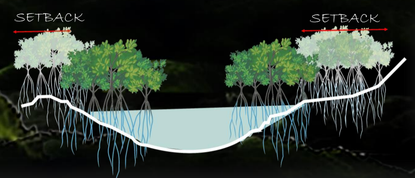
Runoff collected by 13 water catchments eventually ends up at the mouth of Berlayer Creek. The water level in that area is largely dependent on the flood and ebb tides, corresponding to inland flood risk.



Catchment	Area (ha)	Catchment	Area (ha)
C1	1.3	C8	3.4
C2	5.8	C9	1.3
C3	10.9	C10	5.1
C4	2.4	C11	8.2
C5	3.1	C12	9.4
C6	9.0	C13	1.0
C7	9.9		

MANGROVE EXPANSION

The main goal of mangrove setbacks is to preserve biodiversity and the growth of nature. Mangroves can retreat and rise by creating a setback zone with a gradual slope.



Survival depends on sediment supply and reestablishing tidal hydrology. To expand the mangrove habitat, the platform level is reduced to create pathways for water entry during high tide and plant seedlings in the zone.

Creating inland mangroves with an early head start improves their survival and coastal protection role. Urban sediments can collect over time, creating a natural buffer that benefits various ecosystems. Mangroves also provide cultural ecosystem services including education, recreation, and aesthetics when they are accessible to everyone.

To cope with a 5m rise in sea level, deployable flood barriers are employed along the creek. Increasing the barrier's height can be achieved by adding more beams and taller support posts, although this may necessitate additional storage space.

TIDAL POOL

To increase their ecological significance, tidal pools can be constructed on seawalls with intertidal terraces.

Pools improve biodiversity by reducing desiccation and heat stress. Their design can be modified to offer more habitats for different fauna. To enhance their recreational and educational potential, they could be made visible and accessible to the public during low tide.



Black Margined Nudibranch
(Glossodoris atromarginata)



Pacific Turban Snail
(Turbo bruneus)

Eco-concrete rockpools support invertebrates year-round, including sea slug. Establishing pools at different tide levels is crucial to increase species diversity, as some require specific tidal heights while others need a range throughout their life cycles.

